$\boxed{1}$ 17.11.2020

Lineer Programlama Teorisi

$Mak\ z = x_1 + 3x_2$

$3x_1 + 2x_2 \geq 6$

$3x_1 + x_2 = 4$

$x_1, x_2 \geq 0$

a) Verilen Problemi çözünüz.

b) Verilen problemin dualini alınız

c) Dual problemi çözmeden (a) şıkkından faydalanarak dual problemin optimal çözümünü ve optimal amaç fonksiyonunun değerini yazınız.

$Mak\ z = x_1 + 3x_2$

$-3x_1 - 2x_2 \leq -6$

$3x_1 + x_2 = 4$

Duali almak için problem max amaca sahip ise kısıtları küçüğe eşitle (= ise ellenmiyor ona karşılık gelen ~~değ~~ değişken işaretten bağımsız oluyor)

Dualini alır isek

$-3y_1 + 3y_2 \geq 1$

$-2y_1 + y_2 \geq 3$

$y_1 \geq 0$ $y_2$ işaretten bağımsız

$Min\ w = -6y_1 + 4y_2$

Şimdi verilen problemin (a) şıkkını çözelim.

$$\begin{array}{l} Mak\ z = x_1 + 3x_2 \\ 3x_1 + 2x_2 \geq 6 \\ 3x_1 + x_2 = 4 \\ x_1, x_2 \geq 0 \end{array} \Rightarrow \begin{array}{l} Mak\ z = x_1 + 3x_2 \\ 3x_1 + 2x_2 - x_3 = 6 \\ 3x_1 + x_2 = 4 \\ x_1, x_2, x_3 \geq 0 \end{array} \Rightarrow \begin{array}{l} Mak\ z = x_1 + 3x_2 - Mx_4 - Mx_5 \\ 3x_1 + 2x_2 - x_3 + x_4 = 6 \\ 3x_1 + x_2 + x_5 = 4 \\ x_1, \dots, x_5 \geq 0 \end{array}$$

2

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 3 | 0 | $-M$ | $-M$ | · |
| j | $c_j$ | $V_0$ | $V_1$ | $V_2$ | $V_3$ | $V_4$ | $V_5$ | OR |
| 4 | $-M$ | $x_4=6$ | 3 | 2 | $-1$ | 1 | 0 | $6/3$ |
| 5 | $-M$ | $x_5=4$ | 3 | 1 | 0 | 0 | 1 | $4/3$ |
| $c_j-z_j$ | | $-10M$ | $1+6M$ | $3+3M$ | $-M$ | 0 | 0 | |

$\begin{pmatrix} 1 & -1 \\ 0 & 1/3 \end{pmatrix} \leftarrow$

$\uparrow$ TG

$V_1$ Tabana girer $V_5$ Tabandan çıkar.

Yeni Taban $(V_4, V_1) = \begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & 3 & 0 & 1 \end{pmatrix} N \begin{pmatrix} 1 & 3 & 1 & 0 \\ 0 & 1 & 0 & 1/3 \end{pmatrix}$

$N \begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1/3 \end{pmatrix}$

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 3 | 0 | $-M$ | $-M$ | |
| j | $c_j$ | $U_0$ | $V_1$ | $V_2$ | $V_3$ | $V_4$ | $V_5$ | |
| 4 | $-M$ | $x_4=2$ | 0 | 1 | $-1$ | 1 | $-1$ | 2 |
| 1 | 1 | $x_1=4/3$ | 1 | $1/3$ | 0 | 0 | $1/3$ | 4 |
| $c_j-z_j$ | | $-2M+4/3$ | 0 | $8/3+M$ | $-M$ | 0 | $-2M-1/3$ | |

TG $\leftarrow$

$\begin{pmatrix} 1 & 0 \\ -1/3 & 1 \end{pmatrix}$

$\uparrow$ TG

yeni Taban $(V_2, V_1)$ $N \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1/3 & 1 & 0 & 1 \end{pmatrix} N \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1/3 & 1 \end{pmatrix}$

3

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 3 | 0 | $-M$ | $-M$ | |
| $j$ | $c_j$ | $V_0$ | $V_1$ | $V_2$ | $V_3$ | $V_4$ | $V_5$ | |
| 2 | 3 | $x_2=2$ | 0 | 1 | $-1$ | 1 | $-1$ | $\times$ |
| 1 | 1 | $x_1=2/3$ | 1 | 0 | $1/3$ | $-1/3$ | $2/3$ | 2 |
| $c_j-z_j$ | | $z_0=20/3$ | 0 | 0 | $8/3$ | $-M+2/3$ | $-M+7/3$ | |

$\begin{pmatrix} 1 & 3 \\ 0 & 3 \end{pmatrix}$

TÇ ⇐

⇑ TG

Yeni Taban $(V_2, V_3) \sim \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & 1/3 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 3 \end{pmatrix}$

$\sim \begin{pmatrix} 1 & 0 & 1 & 3 \\ 0 & 1 & 0 & 3 \end{pmatrix}$

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 3 | 0 | $-M$ | $-M$ | |
| $j$ | $c_j$ | $V_0$ | $V_1$ | $V_2$ | $V_3$ | $V_4$ | $V_5$ | |
| 2 | 3 | $x_2=4$ | 3 | 1 | 0 | 0 | 1 | |
| 3 | 0 | $x_3=2$ | 3 | 0 | 1 | $-1$ | 2 | |
| $c_j-z_j$ | | $z_0=12$ | $-8$ | 0 | 0 | $-M$ | $-M-3$ | |

$x_1=0 \quad x_2=4 \quad x_3=2 \quad x_4=x_5=0 \quad z_{Max}=12$

$c_4-z_4=-M \Rightarrow -z_4=0 \quad z_4=0 \quad y_1=0$

$c_5-z_5=-M-3 \Rightarrow z_5=3 \quad y_2=3$

$\text{Min } z = x_1+x_2+x_3$

$2x_1+x_2+4x_3=20$

$x_1+2x_2+x_3=30$

$x_1, x_2, x_3 \geq 0$

a) verilen problemi çözünüz

b) Dualını alınız

c) primal tablodan dual problemi çözmeden optimal amacı ve optimal dual çözümünü bul.

$\boxed{4}$

$$\text{Min } z = x_1 + x_2 + x_3 + Mx_4 + Mx_5$$

$$2x_1 + x_2 + 4x_3 + x_4 = 20$$

$$x_1 + 2x_2 + x_3 + x_5 = 30$$

$$x_1, \dots x_5 \geq 0$$

$\begin{pmatrix} 1/4 & 0 \\ -1/4 & 1 \end{pmatrix}$

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 1 | 1 | M | M | |
| | | | $V_1$ | $V_2$ | $V_3$ | $V_4$ | $V_5$ | OR |
| j | $C_j$ | $V_0$ | | | | | | |
| 4 | M | $x_4 = 20$ | 2 | 1 | 4 | 1 | 0 | 5 |
| 5 | M | $x_5 = 30$ | 1 | 2 | 1 | 0 | 1 | 30 |
| $c_j - z_j$ | | $z_0 = 50M$ | 1-3M | 1-3M | 1-5M | 0 | 0 | |

$\uparrow$ TG

$V_4$ çıkar. $V_3$ girer.

$$(V_3, V_5) \sim \begin{pmatrix} 4 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1/4 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 0 & 1/4 & 0 \\ 0 & 1 & -1/4 & 1 \end{pmatrix}$$

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 1 | 1 | M | M | |
| | | | $V_1$ | $V_2$ | $V_3$ | $V_4$ | $V_5$ | |
| j | $C_j$ | $V_0$ | | | | | | |
| 3 | 1 | $x_3 = 5$ | 1/2 | 1/4 | 1 | 1/4 | 0 | 20 |
| 5 | M | $x_5 = 25$ | 1/2 | 7/4 | 0 | -1/4 | 1 | $\frac{100}{7}$ |
| $c_j - z_j$ | | $z_0 = 5 + 25M$ | $\frac{1}{2} - \frac{M}{2}$ | $-7M/4$ | 0 | $\frac{5M}{4} - \frac{1}{4}$ | 0 | |

T.G. $\leftarrow$

$\leftarrow$

$\uparrow$

$\begin{pmatrix} 1 & -1/7 \\ 0 & 4/7 \end{pmatrix}$

5

$V_2$ girer. $V_5$ çıkar.

$$(V_3 V_2) \sim \begin{pmatrix} 1 & 1/4 & 1 & 0 \\ 0 & 7/4 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1/4 & 1 & 0 \\ 0 & 1 & 0 & 4/7 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 0 & 1 & -1/7 \\ 0 & 1 & 0 & 4/7 \end{pmatrix}$$

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | $c_5$ | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 1 | 1 | M | M | |
| $j$ | $c_j$ | $V_0$ | $V_1$ | $V_2$ | $V_3$ | $V_4$ | $V_5$ | OR |
| 3 | 1 | $x_3=\frac{10}{7}$ | $3/7$ | 0 | 1 | $1/7$ | $-1/7$ | |
| 2 | 1 | $x_2=\frac{100}{7}$ | $2/7$ | 1 | 0 | $-1/7$ | $4/7$ | |
| $c_j-z_j$ | | $z_0=\frac{110}{7}$ | $+2/7$ | 0 | 0 | M | $M-3/7$ | |

$x_1=0 \quad x_2=\frac{100}{7} \quad x_3=10/7 \quad z=\frac{110}{7}$

$2y_1+y_2 \le 1$

$y_1+2y_2 \le 1$

$4y_1+y_2 \le 1$

Maks $w = 20y_1+30y_2$

$y_1, y_2$ işaretten bağımsız.

$c_4-z_4=M \Rightarrow z_4=y_1=0$

$c_5-z_5=M-3/7 \quad z_5=y_2=3/7$

$z_0=110/7$

$\boxed{6}$ $\quad$ $\text{Min } z = 4x_1 + 2x_2$

$$x_1 + x_2 = 1$$
$$3x_1 - x_2 \geq 2$$
$$x_1, x_2 \geq 0$$

Dual simpleks algoritması ile çözelim

$$\text{Min } z = 4x_1 + 2x_2 + Mx_3$$
$$x_1 + x_2 + x_3 = 1$$
$$3x_1 - x_2 - x_4 = 2$$

$$\text{Min } z = 4x_1 + 2x_2 + Mx_3$$
$$x_1 + x_2 + x_3 = 1$$
$$-3x_1 + x_2 + x_4 = -2$$
$$x_1, \ldots x_4 \geq 0$$

$\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | |
|---|---|---|---|---|---|---|---|
| | | | 4 | 2 | M | 0 | |
| j | $c_j$ | $v_0$ | $v_1$ | $v_2$ | $v_3$ | $v_4$ | or |
| 3 | M | $x_3 = 1$ | 1 | 1 | 1 | 0 | 1 |
| 4 | 0 | $x_4 = -2$ | -3 | 1 | 0 | 1 | X |
| $c_j - z_j$ | | $z_0 = -M$ | $4-M$ | $2-M$ | 0 | 0 | |

$\uparrow$

$$(v_2, v_4) \sim \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{pmatrix}$$

$\begin{pmatrix} 1 & 1/4 \\ 0 & -1/4 \end{pmatrix}$

T.G $\leftarrow$

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | ~~X~~ | X |
|---|---|---|---|---|---|---|---|---|
| | | | 4 | 2 | M | 0 | | |
| j | $c_j$ | $v_0$ | $v_1$ | $v_2$ | $v_3$ | $v_4$ | OR | |
| 2 | 2 | $x_2 = 1$ | 1 | 1 | 1 | 0 | | |
| 4 | 0 | $x_4 = -3$ | -4 | 0 | -1 | 1 | | |
| $c_j - z_j$ | | $z_0 = 2$ | 2 | 0 | $M-2$ | 0 | | |

$\uparrow$ T.C

7

Yeni Tabanımız

$$(V_2 \ V_1) \sim \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & -4 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1/4 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 0 & 1 & 1/4 \\ 0 & 1 & 0 & -1/4 \end{pmatrix}$$

| B | | C | $c_1$ | $c_2$ | $c_3$ | $c_4$ | |
|---|---|---|---|---|---|---|---|
| | | | 4 | 2 | M | 0 | |
| j | $c_j$ | $V_0$ | $V_1$ | $V_2$ | $V_3$ | $V_4$ | OR |
| 2 | 2 | $x_2 = 1/4$ | 0 | 1 | 3/4 | 1/4 | |
| 1 | 4 | $x_1 = 3/4$ | 1 | 0 | 1/4 | -1/4 | |
| $c_j - z_j$ | | $z_0 = 7/2$ | 0 | 0 | $M - \frac{10}{4}$ | 3/4 | |

Aşağıdaki problemi

$\text{Max } z = 5x_1 + 10x_2$

$x_1 + x_2 \leq 10$

$x_1 - x_2 \geq 1$

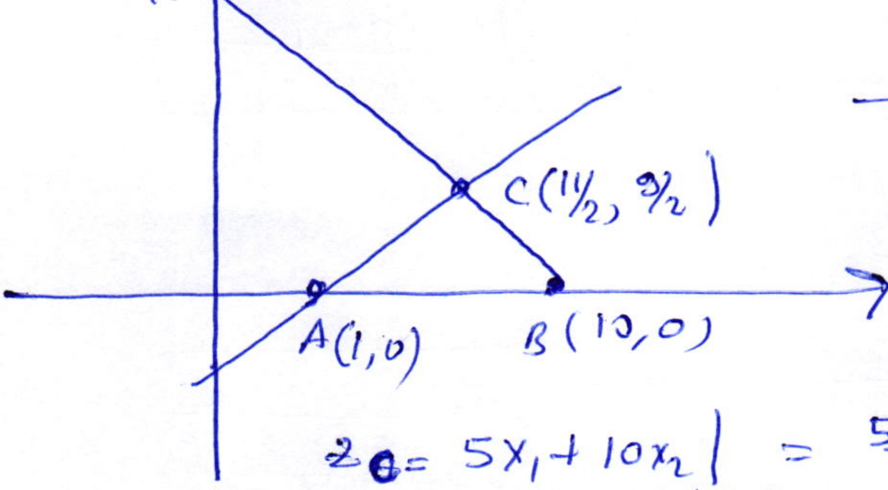
$x_1, x_2 \geq 0$

a) grafik metotla

b) cebirsel metotla

c) bulduğunuz bütün çözümleri uygun, uygun olmayan, temel temel olmayan şeklinde tanımlayın

a) grafik metod.




$x_1 + x_2 = 10$

$x_1 - x_2 = 1$

$2x_1 = 11$

$x_1 = 11/2$

$x_2 = 10 - 11/2$

$x_2 = 9/2$

$z_C = 5x_1 + 10x_2 \Big|_{(11/2,\, 9/2)} = \frac{55}{2} + \frac{90}{2} = \frac{145}{2} = 72.5$

$z_B = 5x_1 + 10x_2 \Big|_{(10,0)} = 50$

$z_A = 5x_1 + 10x_2 \Big|_{(1,0)} = 5$

(C) noktası ($x_1 = 11/2$, $x_2 = 9/2$) $z = 72.5$ optimal.

8

b) cebirsel metod.

$z = 5x_1 + 10x_2$

$x_1 + x_2 + x_3 = 10$

eşitlik haline getirilir.

$x_1 - x_2 - x_4 = 1$

4 bilinmeyen iki denklem (4-2) tane keyfi değişken verilir.

| | | | |
|---|---|---|---|
| $x_1 = x_2 = 0$ | $x_3 = 10$ $x_4 = -1$ | $(0, 0, 10, -1)$ | uygun olmayan temel çözüm |
| $x_1 = x_3 = 0$ | $x_2 = 10$ $x_4 = -11$ | $(0, 10, 0, -11)$ | // |
| $x_1 = x_4 = 0$ | $x_2 = -1$ $x_3 = 11$ | $(0, -1, 11, 0)$ | // |
| $x_2 = x_3 = 0$ | $x_1 = 10$ $x_4 = +9$ | $(10, 0, 0, 9)$ $z^* = 50$ | uygun Temel çözüm |
| $x_2 = x_4 = 0$ | $x_1 = 1$ $x_3 = 9$ | $(1, 0, 9, 0)$ $z^* = 5$ | // |
| $x_3 = x_4 = 0$ | $x_1 = 11/2$ $x_2 = 9/2$ | $(11/2, 9/2, 0, 0)$ $z^* = \frac{145}{2}$ | optimal uygun Temel çözüm |